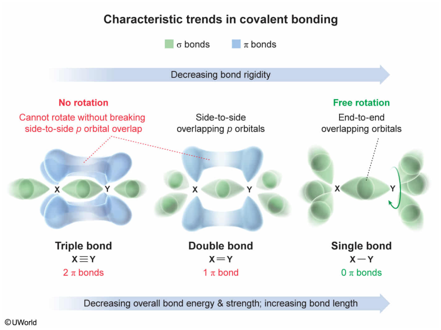
When comparing single, double, and triple covalent bonds formed between two atoms X and Y, decreasing the number of π bonds between the atoms will:

I. decrease the overall bond dissociation energy and bond strength.
II. increase the bond length.
III. decrease the bond rigidity (allow more bond twisting or rotation).

- A. I and II only [6%]
- B. I and III only [7%]
- C. II and III only [8%]
- ✔ D. I, II, and III [77%]

Correct
77% answered correctly

Explanation:



Covalent bonds between atoms are formed by sharing electrons, which is made possible through overlap of the atomic orbitals from each atom. Covalent bonds made by the end-to-end overlap of orbitals are called **sigma (σ) bonds**, whereas covalent bonds made by the side-to-side overlap of *p* orbitals are called **pi (π) bonds**.

Although some atoms participate only in single covalent bonding (by a σ bond), other atoms can participate in **multiple bonds** (ie, double or triple bonds) by the addition of one or more π bonds. In this way, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds. Comparisons can be made between single, double, and triple covalent bonds provided that the bonds being compared involve the same two types of atoms.

The overall dissociation energy of a bond results from the sum of all σ and π bonding contributors (ie, the total energy required to break the σ bond and any π bonds). Therefore, the overall **bond dissociation energy** and **bond strength** tend to **decrease** (relative to a triple bond) as the number of **π bonds decreases (Number I)**.

Because of the shapes of the orbitals involved, **bond length** tends to **increase** (relative to a triple bond) as the number of π bonds decreases **(Number II)**.

Because σ bonds are made by an end-to-end orbital overlap, free rotation around the bond between the two σ-bonded atoms can be achieved while maintaining the orbital overlap. In contrast, the side-to-side overlap of π bonds prevents free rotation (only some limited twisting can occur). Therefore, decreasing the number of π bonds also **decreases the rigidity** (ie, allows more twisting or rotation) of the bond between the atoms **(Number III)**.

Educational objective:
Multiple covalent bonds formed by *s* and *p* orbitals consist of one σ bond (end-to-end orbital overlap) and one or more π bonds (side-to-side *p* orbital overlap). Decreasing the number of π bonds between two atoms increases bond length, decreases bond rigidity, and decreases the overall bond dissociation energy and strength.

Time spent:0 sec
Subject:
General Chemistry

QID:402334
System:
Interactions of Chemical Substances